Juan de Dios Alvarez

Data Scientist -- Machine Learning, NLP and Analytics for Business Decisions

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Profile

Data scientist with seven years of industry work (2019 to present), four of them delivering analytics and machine learning to US-based clients in a distributed, English-speaking team. Master's in Computer Science (machine learning and NLP). I build predictive models, NLP pipelines and dashboards end to end: from data extraction and SQL modeling to model selection by cross-validation, deployment, and reporting that non-technical decision-makers use. I measure every model or method against a baseline or ground truth before it is relied on.

Selected Results

- **NLP to production for a US client:** took a classifier of public school-board minutes from proof of concept to production on Azure ML, feeding a Power BI dashboard that told the client's sales team which products to offer each district.
- **Predictive modeling:** built a crop-production model from satellite vegetation indices at FIRA, a development bank; Random Forest and Gradient Boosting selected by 5-fold cross-validated RMSE and evaluated on held-out data.
- **Data at scale for decision-makers:** replaced an analyst's manual Excel process with 74 scheduled extraction scripts over 9+ sources feeding PostgreSQL, 60+ SQL functions and materialized views, and 15+ dashboards.
- **Measured methods:** compared keyword, embedding and hybrid retrieval by recall@k against a hand-labeled question bank, and LLM extraction against a rule-based baseline on recall, false positives and runtime.

Professional Experience

Data Scientist & ML Engineer | Independent Practice

March 2026 -- present

- Built and run in production a small business's data system: PostgreSQL warehouse (40+ tables), bank-to-invoice-to-payment reconciliation with confidence tiers, bank-statement OCR checked cell by cell against the PDF text layer, and a FastAPI application used daily by staff.
- Clustered short Spanish financial-transaction text without labels (multilingual embeddings, UMAP, HDBSCAN, FINCH), scored by silhouette, DBCV and ARI.
- Designed NLP and LLM pipelines with mechanical checks (schema-constrained outputs, rejection of citations not in the source) and routing of sensitive documents to local models; each call logs latency, tokens and truncation.
- Prototyped an anti-money-laundering compliance assistant: rules-as-data engine with rolling accumulation, thresholds and deadlines, and the inputs and rule version stored behind every alert.

Economic Research Specialist (Data Scientist) | FIRA

January 2025 -- January 2026

- Replaced one analyst's manual Excel process, which fed credit-assignment and non-payment-risk analysis, with a scheduled, unattended pipeline: 74 extraction scripts (Python, aiohttp/asyncio, Playwright) over 9+ sources (INEGI, Banxico, USDA, others) loading PostgreSQL, with 60+ SQL functions and materialized views computing indicators.
- Built a crop-production model from Sentinel-2 and Planet Labs satellite vegetation indices: outlier filtering, PCA, Random Forest and Gradient Boosting selected by 5-fold cross-validated RMSE, evaluated on held-out data.
- Delivered automated executive reports and 15+ Apache Superset dashboards to decision-makers; set governance conventions (raw data preserved, lineage, GitLab) and supervised an undergraduate team.

Data Scientist | CITEIM

October 2024 -- January 2025

- Built the dataset for a Mexican Sign Language recognition system from video: asynchronous, rate-limited bulk download, weak labels from burned-in subtitles (OCR) and audio (offline speech-to-text), and MediaPipe pose and hand keypoints.

Data Scientist & Team Lead | Punto Singular

December 2023 -- August 2024

- Led monthly analytics cycles for an EdTech client: KPI and quality dashboards in R, Python, Tableau and Looker consumed by business stakeholders; Power BI and data-flow architecture for LMS analytics.
- Prototyped real-time WebSocket market-data processing on AWS EC2; founded and led a student Data Science group.

Data Scientist | BairesDev

September 2023 -- November 2023

- Built a retrieval-augmented generation proof of concept that let a legal client's lawyers query their own contracts for obligations, risk clauses and precedents; delivered transformer-based summarization and Neo4j graph modeling.

Data Scientist | D3Clarity (formerly Viqtor Davis North America)

February 2019 -- March 2023

- First data scientist of the company; defined the analytics roadmap for US-based clients in a distributed team.
- Took an NLP classifier of public school-board minutes from proof of concept to production on Azure ML, feeding a Power BI dashboard for the client's sales team.
- Built forecasting models to identify consumer segments, data lakes and ETL on AWS; executive and operational dashboards in Power BI and Tableau for non-technical decision-makers.

Earlier Experience

2002 -- 2019

- Developer and services administrator, University of Michoacan IT (2013 -- 2019); lecturer, Tecnologico de Monterrey Business School; HAWC observatory data analysis (Python, C++); data-acquisition software, CERN Crystal Clear Collaboration (2002).

Education

- **Master's in Computer Science** -- INAOE, Mexico (2006 -- 2009). Degree awarded; SEP professional license (cedula) 5916605. Thesis: automatic text classification using prototype-based class reduction.
- **PhD Candidate in Physics** -- University of Michoacan (2009 -- 2013). HAWC data analysis; co-author on 14 publications.
- **Bachelor's in Physics and Mathematics** -- University of Michoacan (1998 -- 2003).
- **Certifications:** AWS Certified Cloud Practitioner; Semarchy xDM. **Languages:** Spanish (native), English (professional), Italian (basic-intermediate), French (basic).

Technical Skills

- **Machine learning and statistics:** scikit-learn (Random Forest, Gradient Boosting, PCA), pandas, Polars, NumPy, SciPy, clustering (HDBSCAN, FINCH, UMAP); k-fold cross-validation, RMSE/MAE/R2, silhouette/DBCV/ARI; R; PyTorch at prototype level
- **NLP:** text classification, transformer summarization, embeddings (bge-m3, multilingual-e5, sentence-transformers), hybrid retrieval (BM25 + embeddings, RRF), RAG, NER (GLiNER), local LLMs (Ollama), OCR
- **Visualization:** Power BI, Tableau, Looker, Apache Superset
- **Data and SQL:** PostgreSQL (PL/pgSQL, window functions, materialized views, JSONB), SQLite, Oracle, MySQL, Neo4j; ETL, lineage
- **Cloud and engineering:** AWS (EC2, SageMaker), Azure ML, Linux/systemd; Python (asyncio/aiohttp, FastAPI, Flask, pytest), Git (GitHub, GitLab)